SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	M Jeevitha	**Reg. No.:**	192424257
Section / Batch:	2024 Slot C	**Date:**	17-08-2026

Code of Conduct

I **M Jeevitha (192424257)***(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: M Jeevitha

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near…” and “book a flight from Chennai to…”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

1. Constituency Parsing vs Dependency Parsing

Constituency Parsing

Constituency parsing represents a sentence as a hierarchical tree of phrases. It divides a sentence into constituents such as noun phrases (NP), verb phrases (VP), and prepositional phrases (PP).

For example:

"The student read the book."

The structure can be represented as:

Sentence → Noun Phrase + Verb Phrase

- NP → The student
- VP → read + the book
- NP → the book

This representation explains how words are grouped into larger grammatical units. It is useful for identifying phrases and understanding the hierarchical organization of a sentence.

Dependency Parsing

Dependency parsing focuses on direct relationships between individual words. Each word is connected to another word through a dependency relation, with one word generally acting as the head.

For example:

"The student read the book."

- read → student = subject
- read → book = object
- student → The = determiner
- book → the = determiner

Thus, dependency parsing directly shows who performs an action and what the action affects.

Long-Distance Dependencies

Consider:

"The book that the student borrowed was interesting."

Here, "book" is related to "was interesting", even though several words occur between them. Dependency parsing can represent this relationship directly through dependency links.

Constituency parsing can also represent such structures, but the relationship is distributed across different branches of the phrase tree.

Comparison

Feature	Constituency Parsing	Dependency Parsing
Main focus	Phrase structure	Word relationships
Representation	Hierarchical tree	Dependency tree
Units	Phrases/constituents	Individual words
Subject/object identification	Indirect	Direct
Long-distance relations	Can represent	Easily represented directly
Conversational NLP	Useful	Highly useful
Semantic relation extraction	Less direct	More direct

Analysis and Justification

Conversational text often contains short, incomplete, informal, and complex sentences. A conversational AI needs to identify relationships such as who did what, to whom, when, and where.

Dependency parsing is therefore more suitable when the objective is to extract meaningful relationships between words. Constituency parsing remains valuable when the application requires detailed information about phrase boundaries and hierarchical structure.

Conclusion

Dependency parsing is generally more suitable for conversational relationship extraction, while constituency parsing is preferable when detailed phrase-level grammatical structure is required. A practical NLP system can also combine both representations for better syntactic understanding.

2. Top-Down Parsing vs Earley Parsing

Top-Down Parsing

Top-down parsing starts with the start symbol of the grammar and attempts to derive the input sentence by applying grammar rules.

For example:

S → NP VP

The parser first predicts an NP and then a VP before matching them with the input words.

It is relatively simple and easy to understand. However, conventional top-down parsing can have problems when the input is ambiguous or incomplete.

Problem with Incomplete Input

Consider:

“Best hotels near…”

The user has not completed the query. A traditional parser may expect additional words required to complete a grammatical structure.

Similarly:

“Book a flight from Chennai to…”

The parser knows that **“to”** may introduce a destination, but the destination has not yet been entered.

An interactive NLP system must therefore continue processing without assuming that the input is complete.

Earley Parsing

Earley parsing is a dynamic programming parsing algorithm for context-free grammars. It maintains a collection of possible parsing states rather than committing immediately to one interpretation.

It performs three major operations:

1. **Prediction** – predicts possible grammar rules.
2. **Scanning** – matches the next input word.
3. **Completion** – completes previously started grammar rules.

This makes it capable of handling partial and ambiguous input.

Ambiguous Input

Consider:

“I saw the man with a telescope.”

This may mean:

1. I used a telescope to see the man.
2. The man had a telescope.

An Earley parser can maintain both possible structures until more information becomes available.

Comparison

Feature	**Top-Down Parsing**	**Earley Parsing**
Starting point	Start symbol	Maintains parsing states
Complete input	Works well	Works well
Incomplete input	Less flexible	More flexible
Ambiguous input	May require backtracking	Handles alternatives
Complexity	Simple	More computationally complex
Interactive NLP	Limited	More suitable
Grammar support	Usually restricted depending on implementation	General CFG

Analysis and Justification

Interactive systems such as search engines, chatbots, and voice assistants must process input while the user is still typing or speaking. They cannot always wait for a complete sentence.
Earley parsing is better because it can preserve multiple possible interpretations and continue parsing as new words arrive.

Conclusion

Although top-down parsing is simpler, Earley parsing is more appropriate for interactive NLP applications because it provides greater flexibility for incomplete, ambiguous, and complex input.

3. CFG vs PCFG vs Neural Constituency Parsers

Context-Free Grammar (CFG)

A CFG consists of grammar rules that describe how linguistic units can be combined.

For example:

S→NP VP

NP → Det N

VP → V NP

CFG provides a clear and interpretable representation of syntactic structure.
However, natural language frequently contains structural ambiguity, where the same sentence can have multiple parse trees.

Example:

“I saw the man with a telescope.”

The phrase **“with a telescope”** can modify either the verb phrase or the noun phrase.
A basic CFG can generate both interpretations but does not necessarily determine which interpretation is correct.

Probabilistic CFG (PCFG)

A PCFG extends CFG by assigning a probability to each grammar rule.

For example:

- S → NP VP = 0.90
- VP → V NP = 0.70
- VP → VP PP = 0.30

When multiple parse trees are possible, the parser calculates their probabilities and chooses the most probable interpretation.
Therefore, PCFG provides a mechanism for resolving ambiguity.

Advantages of PCFG

- Maintains understandable grammar rules.
- Can rank alternative parse trees.
- Provides probabilistic interpretation.
- More practical than plain CFG for ambiguous sentences.

Limitations of PCFG

Its performance depends heavily on the quality of the grammar and training data. It may also have difficulty representing complex contextual information.

Neural Constituency Parser

Neural constituency parsers use machine learning and neural networks to learn syntactic patterns from large amounts of annotated data.

They can learn:

- Phrase structures
- Contextual relationships
- Complex syntactic patterns
- Long-distance dependencies
- Language-specific patterns

Modern neural parsers can achieve high accuracy because they learn patterns that are difficult to manually encode in grammar rules.

However, their major limitation is interpretability. It may be difficult to explain exactly why a neural model selected one parse tree over another.

Comparison

Feature	**CFG**	**PCFG**	**Neural Parser**
Grammar rules	Explicit	Explicit	Learned
Ambiguity	Represents alternatives	Ranks alternatives	Learns likely interpretation
Interpretability	Very high	High	Lower
Accuracy	Limited	Moderate	Generally high
Data requirement	Low	Training data useful	Large datasets
Flexibility	Limited	Moderate	High
Practical NLP	Basic	Good	Excellent

Analysis and Justification

For a grammar-checking system, interpretability is important because users may want to know why a sentence is considered incorrect.

A pure neural parser can provide strong accuracy but may not explain its decision easily. CFG is easy to interpret but does not effectively rank ambiguous structures.

PCFG provides a good balance because it retains explicit grammar rules while using probabilities to select likely interpretations.

Conclusion

For applications where both explainability and practical ambiguity resolution are important, PCFG is a strong choice. Neural constituency parsers are preferable when maximum parsing accuracy is the primary goal.

4. Feature Structures vs Traditional CFG

Traditional CFG

A traditional CFG describes grammatical structure using rules such as:

S → NP VP

However, these rules do not explicitly specify whether the subject is singular or plural, which person it represents, or what gender it has.

For example:

“The boy plays.”

The CFG can recognize the structure:

NP + VP

but basic CFG does not automatically enforce subject-verb agreement.

Feature Structures

Feature structures extend grammar rules by associating linguistic units with additional properties.

For example:

NP

- Number = Singular
- Person = Third
- Gender = Masculine

Verb

- Number = Singular
- Person = Third
- Tense = Present

The parser can compare these features and ensure that they are compatible.

Feature Unification

Feature structures often use unification.

For example:

Subject: [Number = Singular]

Verb: [Number = Singular]

Since both values match, the structures can be unified.

But:

Subject: [Number = Singular]

Verb: [Number = Plural]

creates a conflict, so the sentence can be rejected or marked as grammatically inconsistent.

More Features

Feature structures can represent:

- Number
- Gender
- Person
- Tense
- Case
- Mood
- Voice
- Definiteness
- Agreement

Importance in Multilingual NLP

Multilingual systems face additional challenges because languages use different grammatical systems.

For example, some languages have extensive gender and case agreement, while others rely heavily on tense, aspect, or person marking.

Using feature structures allows the parser to represent these properties systematically.

Instead of creating many separate grammar rules, a system can use general rules with feature constraints.

Comparison

Feature	**CFG**	**Feature Structures**
Basic syntax	✓	✓
Number agreement	Limited	✓
Gender agreement	Limited	✓

Person agreement	Limited	✓
Tense information	Limited	✓
Multilingual flexibility	Lower	Higher
Grammar constraints	Basic	Detailed

Analysis and Justification

Feature structures make grammar more expressive because they combine syntactic structure with grammatical attributes.

They are especially useful for multilingual NLP because the same basic grammar rule can be applied with different feature values.

Conclusion

Feature structures are more powerful than basic CFG representations for multilingual NLP. They help enforce agreement, tense, gender, person, and other grammatical constraints, improving both syntactic accuracy and language understanding.

5. Subcategorization Frames

Meaning

A subcategorization frame describes the number and types of arguments that a verb expects.

Verbs do not all behave in the same way. Some verbs require no object, while others require one or more complements.

For example:

Intransitive Verb

“John sleeps.”

The verb **sleep** does not require a direct object.

Frame:

NP + V

Transitive Verb

“John opened the door.”

The verb **open** requires a subject and a direct object.

Frame:

NP + V + NP

Ditransitive Verb

“John gave Mary a book.”

The verb **give** can take:

- Subject
- Recipient
- Object

Frame:

NP + V + NP + NP

Another form is:

“John gave the book to Mary.”

NP + V + NP + PP

Verb with Location Requirement

“Put the file on the desk.”

The verb put normally requires a location or destination.

Frame:

NP + V + NP + PP

The sentence:

“John put the file.”

is syntactically incomplete in its usual interpretation because the required location is missing.

Why It Is Important

Subcategorization helps an NLP system identify **which arguments belong to a verb**.

For a voice assistant:

“Give the book to John.”

The system can identify:

- **Give** → action
- **book** → object
- **John** → recipient

For:

“Put the file on the desk.”

it can identify:

- **put** → action
- **file** → object
- **desk** → destination/location

Syntactic Analysis

Subcategorization helps determine whether a sentence follows the expected grammatical pattern of a verb.

For example:

“John sleeps the book.”

is structurally inappropriate because **sleep** does not normally take a direct object.

Semantic Analysis

Subcategorization also supports semantic interpretation because it helps identify the **semantic roles** associated with verb arguments.

For example:

“Sarah gave the book to Ravi.”

- Sarah → **Agent/Giver**
- book → **Theme**
- Ravi → **Recipient**

Thus, subcategorization creates a bridge between syntax and semantics.

Comparison

Verb	Type	Example	Required Structure
Sleep	Intransitive	John sleeps	NP + V
Open	Transitive	John opened the door	NP + V + NP
Give	Ditransitive	John gave Mary a book	NP + V + NP + NP
Put	Object + location	Put file on desk	NP + V + NP + PP

Analysis and Justification

A voice assistant needs to understand not just individual words but also the **requirements of the main verb**. Subcategorization frames allow the system to predict what types of arguments should occur and how they should be interpreted.

This improves:

- Command understanding
- Semantic role labeling
- Grammar checking
- Information extraction
- Question answering
- Voice assistant accuracy

Conclusion

Subcategorization frames are essential for syntactic and semantic analysis because they specify the argument structure of verbs. They help NLP systems determine whether a sentence is grammatically complete and identify the semantic roles of its components.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1	4	4	4	4	4	20	92
2	4	4	4	3	3	18	
3	4	4	4	3	3	18	
4	3	3	4	4	4	18	
5	4	4	3	4	3	18	

Faculty In-charge	Course Coordinator	Program Director
Dr. Kumaragurubaran		
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________